

Understanding Retail Consumer Behavior: Predictors of Spending, Product Choice, and Customer Satisfaction

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Motivation section (Iris)

What is the quality of these elements? Does the title tell us what to expect of the report to follow? Does the motivation section convince the reader that the broader study of this data set could elicit interesting insights?

Description of Dataset (Alice)

As stated, the dataset describes the shopping behavior of consumers, where each row represents an individual customer and their purchasing activity, including what they bought, how much they spent, how often they shop, their demographic attributes, and whether they used discounts or promotional codes.

Generally, the dataset is of good quality with no missing values or duplicate rows. There are 3900 rows in total, and each row corresponds to a unique customer's purchase transaction collected across a variety of U.S. states.

Where does it come from?? As stated, the dataset describes the shopping behavior of consumers, where [insert]

The variables in the dataset are:

- **Customer.ID:** The unique ID of each customer
- **Age:** The age of customer (in years). Range from 18 to 70 years old.
- **Gender:** Gender of customer
- **Item.Purchased:** Type of item that the customer bought, such as Blouse, Sweater, Scarf, etc.
- **Category:** Broader category of the item purchased. There are 4 categories: Clothing, Footwear, Outerwear, and Accessories
- **Purchase.Amount..USD:** The amount spent on the purchased item, in US dollars
- **Location:** The US state where the customer resides
- **Size:** Size of the clothing item. There are 4 sizes: S, M, L, XL
- **Color:** Color of the clothing item
- **Season:**
- **Review.Rating**
- **Subscription.Status**
- **Shipping.Type:** The method of shipping used for the order (e.g., "Express", "Free Shipping", "Next Day Air", "2-Day Shipping", "Store Pickup"). Indicates speed and delivery preferences
- **Discount.Applied:** Whether a discount was applied to the purchase
- **Promo.Code.Used:** Whether a promotion code was applied to the purchase
- **Previous.Purchases:** The number of purchases the customer made before this transaction
- **Payment.Method:** The payment method used for the transaction. There are 6 options: Cash, Bank Transfer, Debit Card, Credit Card, Venmo, and Paypal

- **Frequency.of.Purchases:** How often the customer shops. The frequencies include Weekly, Fort-nightly, Bi-weekly, Monthly, Every 3 Months, Quarterly, and Annually

3 Research Questions

Stated below are our 3 research questions:

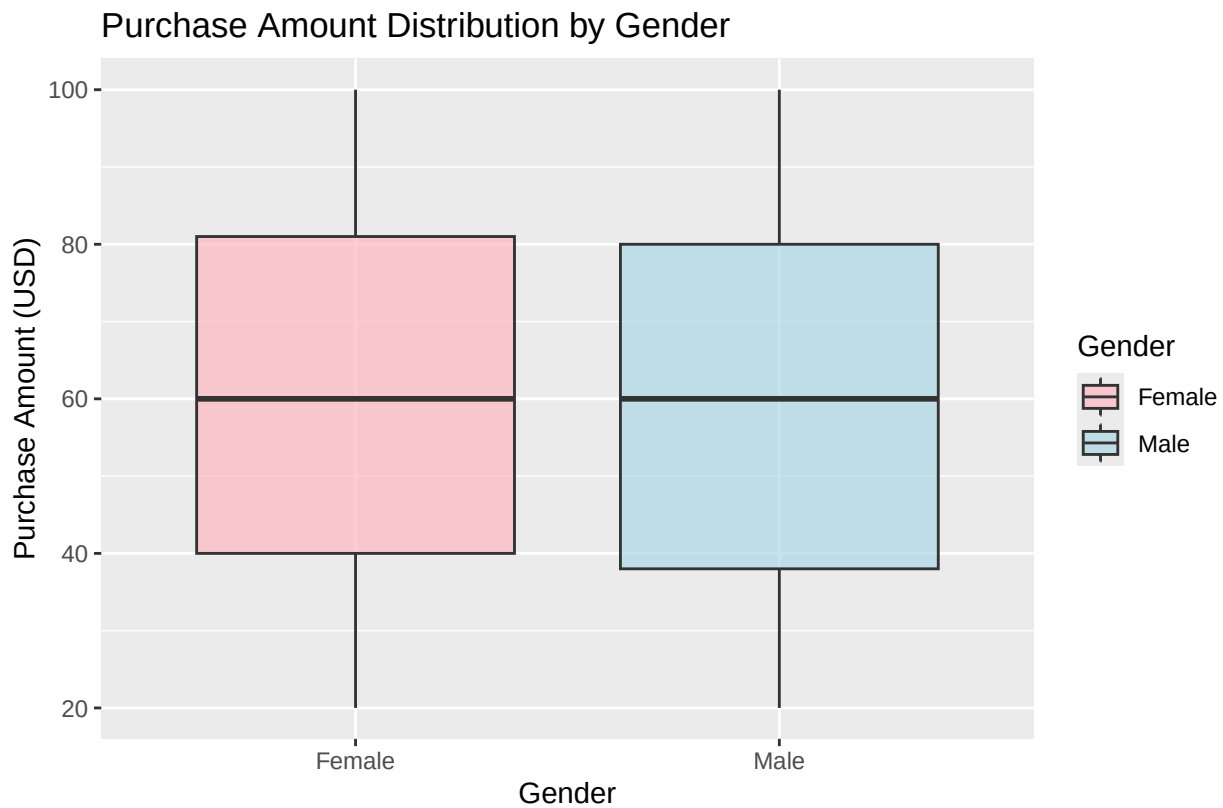
1. What factors predict how much a customer spends (Purchase Amount)?
2. How does season affect the types of items purchased and the color of the items?
3. How do discounts and shipping speed affect customer satisfaction (Review Rating)?

These research questions address core mechanisms of consumer behavior: spending determinants, seasonal purchasing patterns, and satisfaction drivers. Together, they allow us to examine how economic, temporal, and operational factors shape shopping outcomes in a way that is meaningful for both behavioral analysis and retail decision-making.

Question 1: What factors predict how much a customer spends (Purchase Amount)?

Boxplots - Hailey

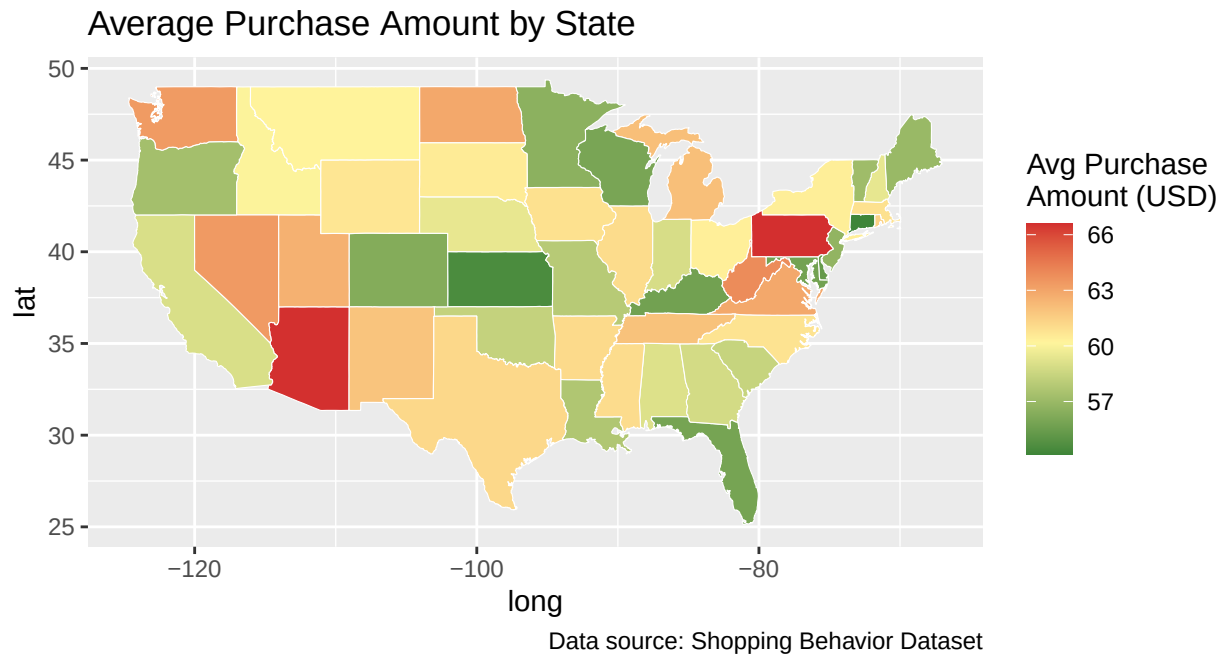
Purchase Amount vs. Gender



Data source: Shopping Behavior Dataset

Spatial Graph - Hailey

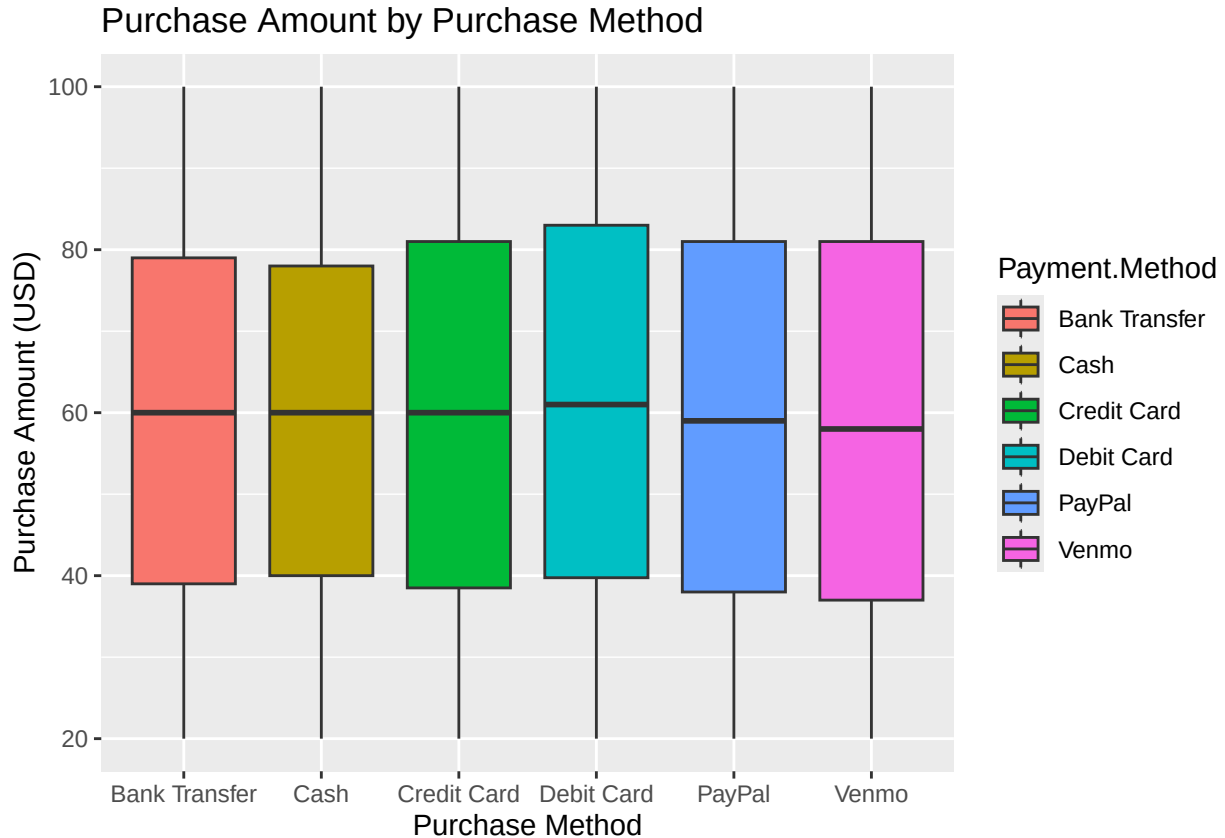
Average Purchase Amount by Location



Boxplot - Iris

Correlations between:

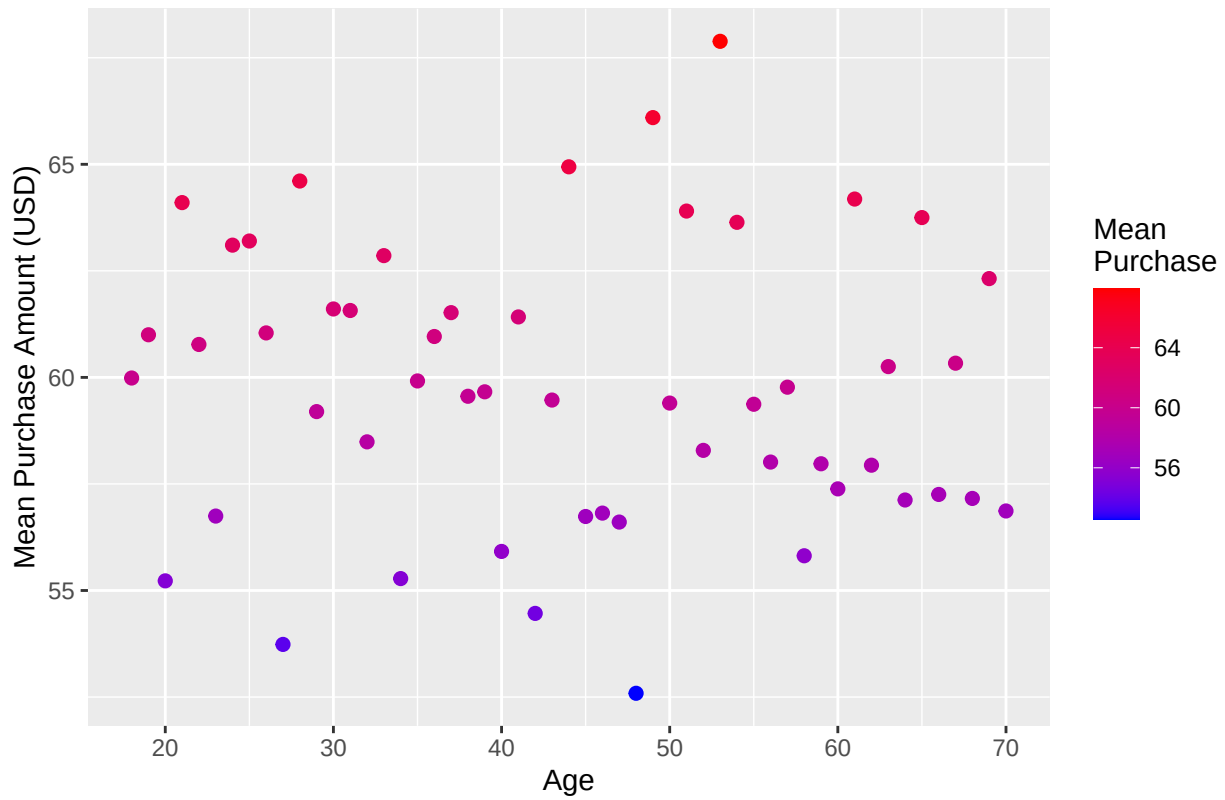
Purchase Amount Purchase method



Scatterplot - Iris

Purchase Amount vs Age

Average Purchase Amount by Age



Question 2: How does season affect the types of items purchased and the color of the items?

Season does not seem to have a significant affect on the types of items purchased and color of the items.

Variables involved: Season (Winter, Spring, Summer, Fall).Type of item purchased – e.g., Category or Item Purchased, Color of item – Color

Possible analyses:Use visualizations such as stacked/clustered bar charts or mosaic plots to compare item types and colors across seasons.

Chi-square test for association

Logistic regression (Predict “Promo Code Used = Yes/No”)

Word Cloud - Michelle

color frequency by season



Heatmap - Michelle

This handles large numbers of categories beautifully.

Why it works:

Items go on the Y-axis

Seasons go on the X-axis

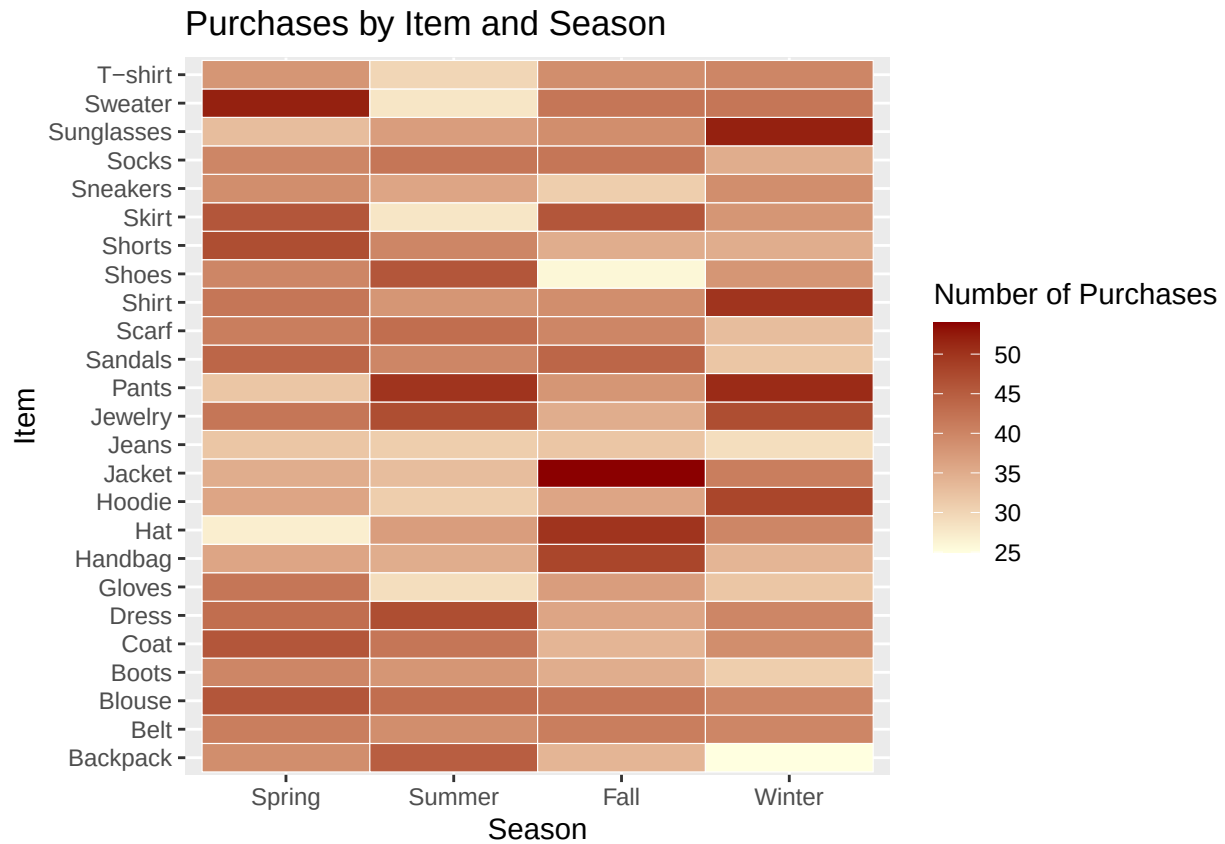
Color = number of purchases

It scales well even if you have 20–50+ item types.

Interpretation:

Darker = purchased more often

Lighter = purchased less often Frequency of Purchases (colored by subscription status) vs. purchase amount
 If frequency is truly numeric and continuous, this can show trends.



1. Chi-Square Test of Independence (Most appropriate)

Use this when both variables are categorical.

You can run: a. Season $\times$ Item Purchased

Tests whether certain items are bought more often in some seasons.

b. Season $\times$ Color

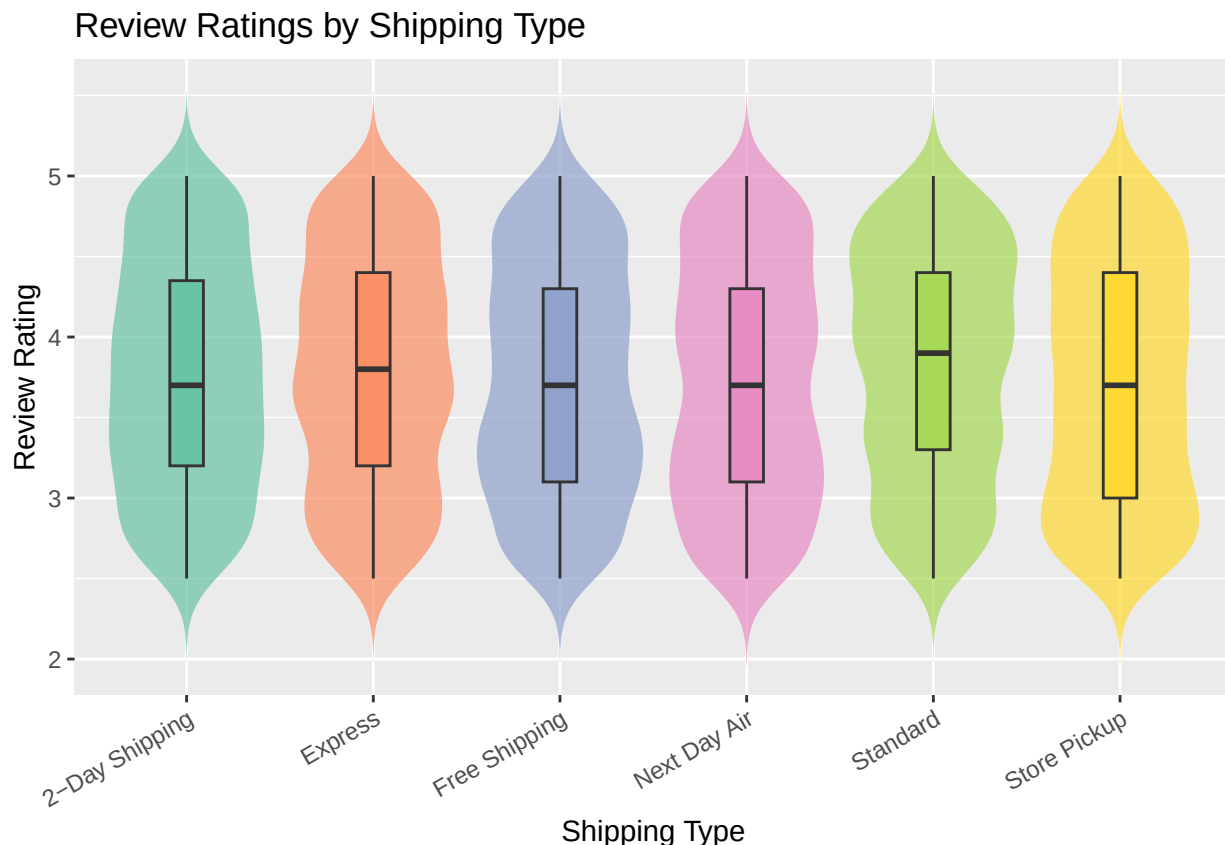
Tests whether certain colors are bought more often in certain seasons.

Question 3: How do shipping speed and discount affect customer satisfaction (Review Rating)?

Shipping speed and discounts are key components of the customer experience, and both may meaningfully influence how satisfied customers feel with their purchase. Hence, we would visualize the distribution of customer satisfaction across different shipping methods and whether there are discount applied using violin plots. Besides providing the the 5-number summary like what a box plot would, a violin plot also shows the distribution so we can examine whether there is skewness or outliers as well.

Then, we would conduct an ANOVA test to examine the mean review ratings across 6 different shipping methods and a two-sample t-test to compare review ratings between two groups (“Yes” vs. “No”). These tests would help us answer whether in this dataset, shipping speed and discount have a statistically significant relationship with customer satisfaction.

Review Raings by Shipping Type



From the following violin plot, we observe that the median rating are fairly similar across all shipping types, ranging between 3.6 to 3.7. Yet, standard and express shipping have a slightly higher median. The other values like min, max, first and third quartile seem to be similar as well. There's no outlier and skewness for any shipping type. All shipping types exhibit a similar spread, which suggests that variability in satisfaction may not strongly be tied to delivery method.

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## Shipping.Type    5      6  1.2048   2.353 0.0384 *
## Residuals 3894 1994  0.5121
## ---
```



```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Because the p-value for the ANOVA test is smaller than 0.05, we can conclude that average satisfaction differs across at least some shipping categories. However, when accompanying with the graph above, we see that there's not much variation in the distribution of average rating across shipping types. Hence, we can say that while shipping type may have a statistically detectable association with review ratings, the practical differences between shipping methods are likely minimal.

Review Raings vs. Discount Applied



The violin plot comparing review ratings for customers who did and did not receive a discount shows that the two groups have nearly identical distributions. Both groups center around ratings of approximately 3.7, with very similar spreads and shapes. There is no visible indication that receiving a discount leads to higher or lower satisfaction. The substantial overlap between the groups suggests little practical difference.

```
##
##  Welch Two Sample t-test
##
## data:  Review.Rating by Discount.Applied
## t = 0.77934, df = 3604.9, p-value = 0.4358
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## -0.02737548  0.06349679
## sample estimates:
## mean in group No mean in group Yes
##      3.757715      3.739654
```

The difference in mean ratings between customers who did not receive a discount (mean = 3.758) and those who did (mean = 3.740) is not statistically significant ($t = 0.779$, $p = 0.436$). The 95% confidence interval for the mean difference includes a zero (-0.027 to 0.063), indicating that any difference is small and likely due to random variation. Overall, the evidence suggests that discount usage does not meaningfully influence customer satisfaction.

From this dataset, we can conclude that the usage of discount does not affect rating. Even though the ANOVA test shows that there is statistical significance between shipping type and rating, the violin plots suggest there is not much variation.

Evaluation/Future work (Michelle)